INDEX

NAME SAIFUDEEN.S SUBJECT COMPUTER NETWORKS OBSERVATION

STD. IT DIV. D ROLL NO. 241001241 SCHOOL REC

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completed

Exp no: 1
Date: 17-07-26

BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS OS

AIM:

To study and understand basic networking commands used in Linux and windows OS.

COMMANDS OBSERVED:

S. No	Command	Linux	Windows
1.	Display IP config	ip config or ip addr	ipconfig
2.	Check network connectivity	ping <host>	ping <host>
3.	Trace route	traceroute <host>	tracert <host>
4.	Display routing table	route -n or ip route	route print
5.	Display network connections	netstat -a	netstat -a
6.	DNS lookup	nslookup <domain>	nslookup <domain>
7.	Display ARP cache	arp -a	arp -a
8.	Display hostname	hostname	hostname
9.	SS - tuln	SS - tuln	netstat -ano
10.	Dig DNS lookup	dig <domain>	nslookup <domain>
11.	DNS information	dig	nslookup
12.	Host lookup	host	nslookup
13.	Socket statistics	SS	netstat -a no
14.	Download web content	curl	curl
15.	Download files	wget	curl -o
16.	Pocket capture	tcpdump	pktmon
17.	network diagnostics	mtr	pathping
18.	network management	nmcli	netsh
19.	network adapter info	ethtool	getmac
20.	Wireless configuration	iwconfig	netsh wlan show interface

BASIC NETWORKING COMMANDS

OUTPUTS

S.NO	OUTPUT
1.	IPV4 Address 192.168.1.00
2.	Reply from 142.250.183.110 : bytes = 32 time = 18ms TTL = 117
3.	Trace complete : Total hops : 10
4.	0.0.0.0.0.0.0.0 192.168.1.1 192.168.1.100
5.	TCP 0.0.0.0:80 0.0.0.0:0 LISTENING
6.	Name : example.com Address : 93.184.216.34
7.	192.168.1.100 - 1a - 2b - 3c - 4d - 5e dynamic
8.	DESKTOP - 7HK82M4
9.	TCP 127.0.0.1 : 49735 0.0.0.0:0 LISTENING
10.	Server : dns.google Address : 8.8.8.8
11.	Default server : dns.google
12.	TCP 128.0.0.1.47385.0.0.0:0
13.	Default : dns.google
14.	<!DOCTYPE html><html : ....>
15.	1001256 100 1256 00 1256.-- : -- : -- : -- : 1256
16.	pktmon successfully started
17.	Computing statistics for 100 seconds
18.	The following commands are available : adv firewall bridge dncp dnsclient interface
19.	00-15-50-01-02-03/Device/TCPIP-{xxxxx-xxx-xxxxx-xxxxxx}
20.	State : connected

Result:

The basic networking commands in linux and windows were successfully studied and observed.

OUTPUT:

IPV4 Address 192.168.1.10

Subnet mask 255.255.255.0

Default gateway 192.168.1.1

DNS Server 8.8.8.8

Exp no: 2
Date: 17/07/26

LEARNING AND ASSIGNMENT OF IP ADDRESS MANUALLY TO COMPUTERS

AIM:

To Learn how manually assign an IP address, subnet mask, default gateway and DNS server to computer and verify network configuration.

PROCEDURE:

1. Open control panel → network and sharing center
2. Select change adapter settings
3. Right click active network adapter and select properties
4. Open internal Protocol version 4 (TCP/IPV4)
5. Select and use following IP address
6. Enter following values

   IP-Address: 192.168.1.10
   Subnet mask: 255.255.255.0
   Default gateway: 192.168.1.1
   Preferred DNS server: 8.8.8.8

7. Click ok and save settings
8. Verify the configuration using ipconfig command

RESULT:

Thus, the IP address was successfully assigned manually and the system displayed and the network configuration was verified using ipconfig command

TYPES OF NETWORK CABLES

Choose the Right Cable for Reliable & High-Performance Networks

1 UNSHIELDED TWISTED PAIR (UTP) CABLES



Most common cable for LAN, Ethernet, and office networks.

STRUCTURE

- Outer Jacket
- Twisted Pairs

KEY BENEFITS

- Commonly used in homes and offices
- Flexible, affordable, and easy to use
- For Ethernet and VoIP applications

2 SHIELDED TWISTED PAIR (STP) CABLES



Ideal for environments with high EMI and interference.

STRUCTURE

- Outer Jacket
- Foil Shielding
- Twisted Pairs

KEY BENEFITS

- Protect against interference and crosstalk
- Require proper grounding
- Ideal for environments with heavy EMI such as factories and industrial setups

3 COAXIAL CABLES



Used for traditional broadband, CCTV, and RF transmissions.

STRUCTURE

- Outer Protective Jacket
- Metallic Shield
- Copper Conductor

KEY BENEFITS

- Highly resistant to signal obstruction
- Great for long-distance transmissions
- Commonly used in data centers, ISP networks, and radio/broadcast applications

4 FIBER OPTIC CABLES



Built for speed and performance in modern networks.

STRUCTURE

- Outer Jacket
- Glass Fiber Strands
- Light Signal Transmission

KEY BENEFITS

- Immune to electromagnetic interference
- Ideal for ultra-fast, long-range data transmission
- Widely used in data centers, ISP networks, and high-bandwidth applications

Tip: Selecting the right cable improves network performance, reduces downtime, and ensures a reliable network infrastructure.

Exp no: 3
Date: 23/07/26

STUDY OF DIFFERENT TYPES OF NETWORK CABLES AND CRIMPING OF CABLE WITH RJ45 CONNECTOR

AIM:

To study different types of network cables and perform crimping of UTP cable using RJ45 connector.

MATERIALS REQUIRED:

- UTP (Cat 5e / Cat6) cable
- RJ45 connector
- Crimping tools
- Wire stripper
- Cable cutter
- LAN cable tester

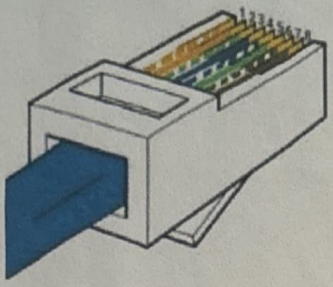
TYPES OF NETWORK CABLES:

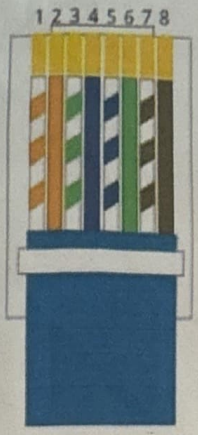
i. UTP (Unshielded Twisted Pair):
   - Most commonly used for ethernet network

ii. STP (Shielded Twisted pair):
   - Has shielding to reduce electromagnetic interference

iii. Coaxial cable:
   - Used for cable TV and older networking systems

iv. Fibre optic cable:
   - Uses light to transmit data at very high speeds over long distances





1. White Orange	5. White Blue
2. Orange	6. Green
3. White Green	7. White Brown
4. Blue	8. Brown

CRIMPING PROCEDURES:

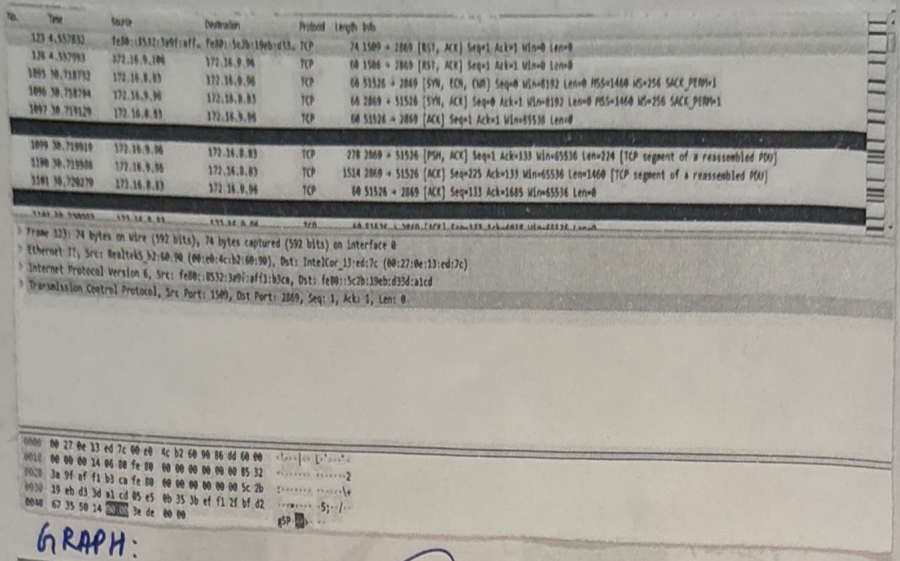
1. Cut the cable to required length
2. Strip about 2-3cm of outer insulation
3. Untwist and arrange the wires according to T568B standard:
   - White/Orange
   - Orange
   - White/Green
   - Blue
   - White/Blue
   - Green
   - White/Brown
   - Brown
4. Trim wires evenly
5. Insert wires fully into RJ45 connector
6. Use crimping tool to crimp connector
7. Test the cable using LAN cable tester

RESULT:

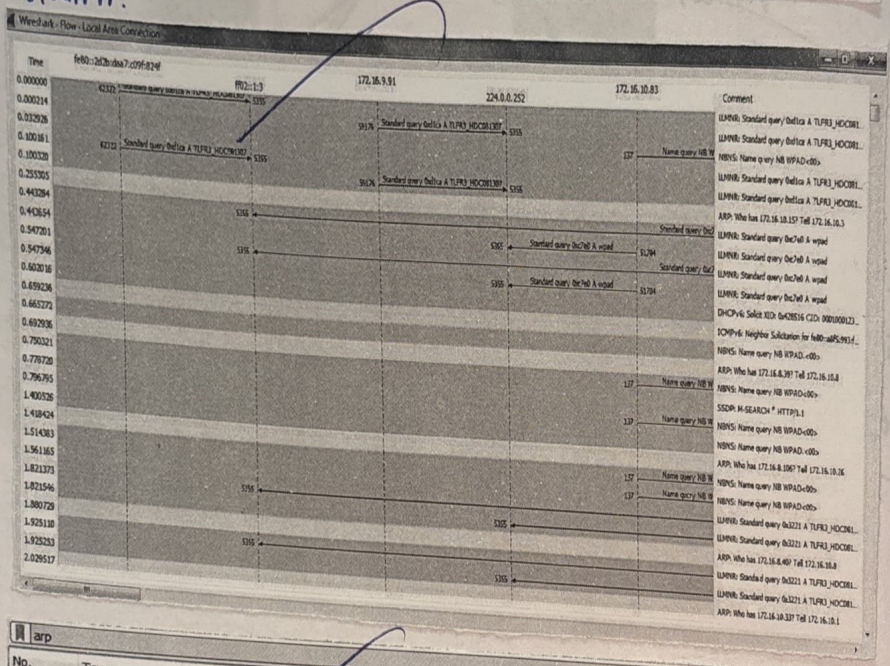
The network cable was successfully crimped with an RJ45 connector and tested for proper connectivity.

OUTPUTS:

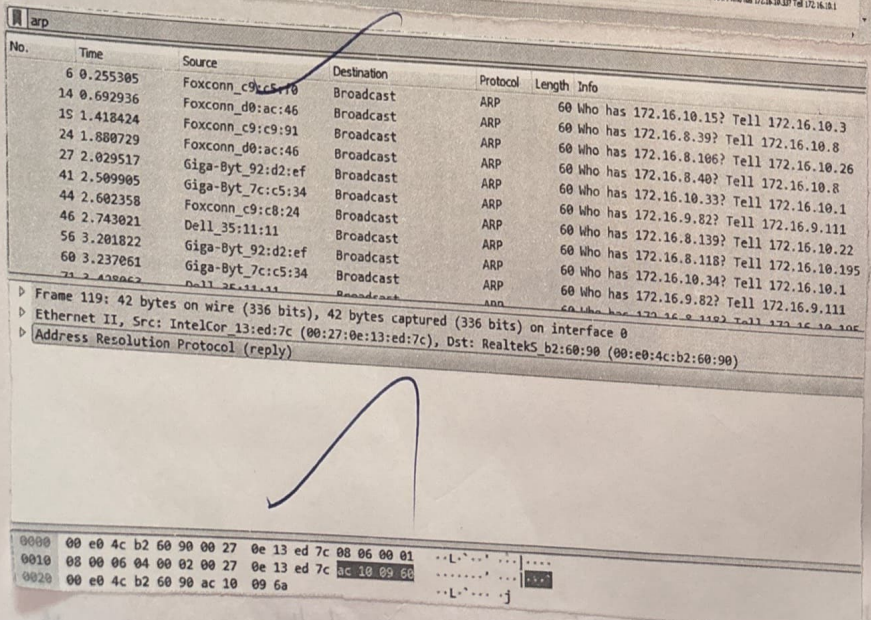
1.



GRAPH:



2.



Exp no: 4

Date: 24-07-2026

IMPLEMENT PACKET SNIFFING IN WIRESHARK

AIM:

To capture and analyse various network protocols packets (like TCP, UDP, ARP, DNS, HTTP, ICMP, DHCP) using wireshark and apply display filters to inspect them.

SOFTWARE USED:

Wireshark Network Protocol Analyzer

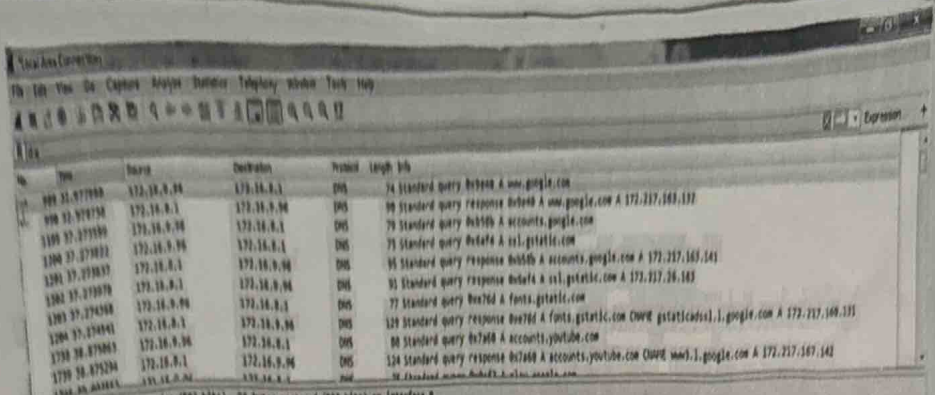
PROCEDURE:

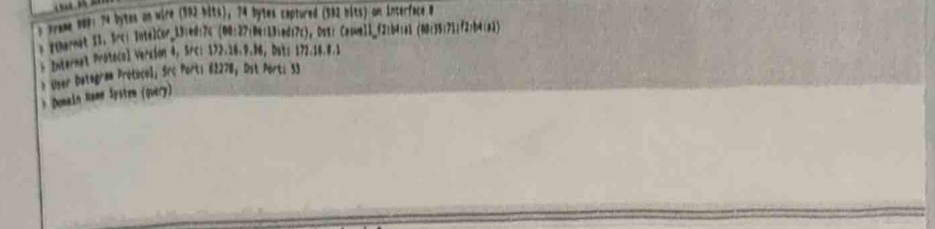
1. Open wireshark
2. Select local area connection (or your active interface)
3. Go to capture → options
4. Set: Stop capture automatically after 100 packets
5. Click start to begin capturing
6. Use filter bar to type specific protocols like TCP, ARP, etc.
7. Use statistics → Flow graph (for TCP/DNS) to view packet flow
8. Click file → save to store captured packets

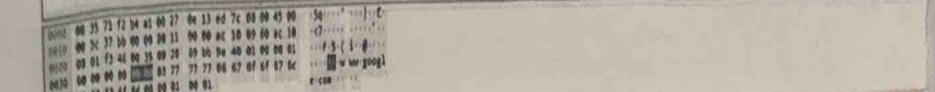
TASKS & OUTPUT:

1. Filter: Display TCP/UDP packets and flow graph
   - Filter used: TCP or UDP
   - Output: Displayed only TCP/UDP packets
   - Flow graph: Viewable from statistics → Flow graph
2. Filter: Display only ARP packets
   - Filter used: ARP
   - Output: Displayed only TCP/UDP packets
   - Flow graph: Viewable from statistics → Flow graph

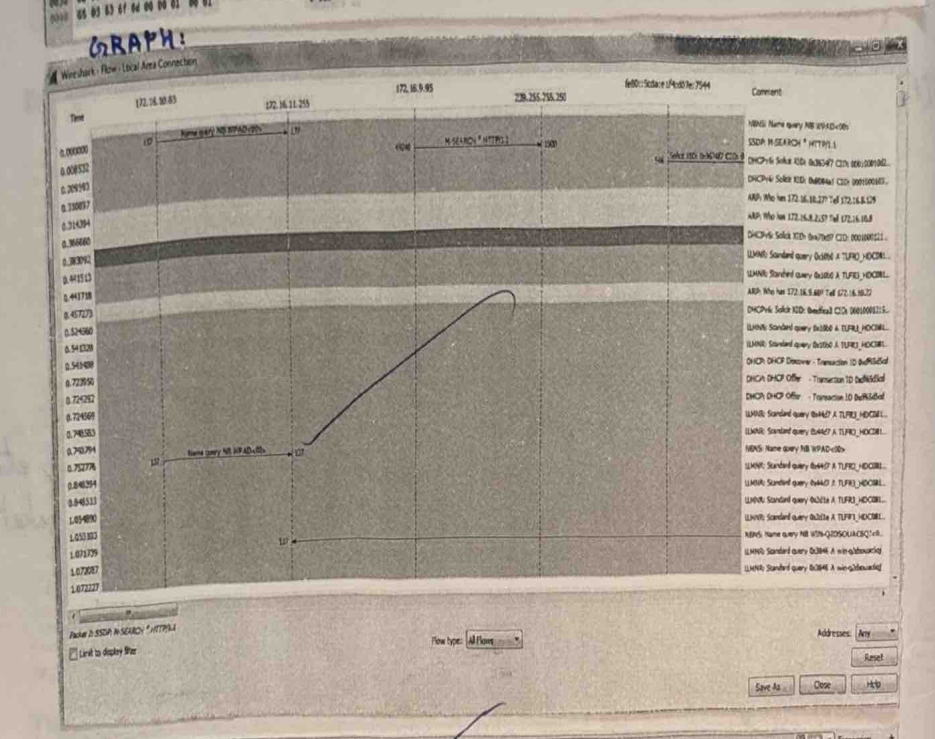
3.



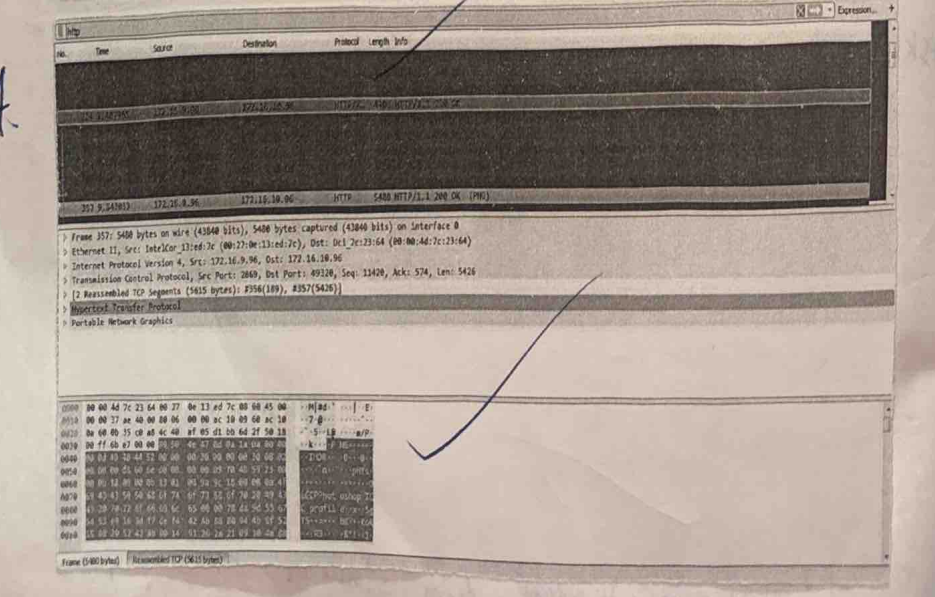




GRAPH:



4.



3. Filter: Display DNS packets and Flow graph
   - Filters used: dns
   - Output: Displayed only DNS queries and responses
   - Flow graph: Shows DNS request - response communication
4. Filter: Display only HTTP packets
   - Filters used: http
   - Output: Captured HTTP traffic (browser or server communication)
5. Filter: Display IP/ICMP packets
   - Filters used: icmp or ip
   - Output: Shows ICMP packets (used in ping operations) and IP packet info
6. Filter: Display only DHCP packets
   - Filters used: bootp (used for DHCP)
   - Output: Displays DHCP discover, offer, request, acknowledgement packets

RESULT:
Successfully captured and filtered different protocols using Wireshark and understood how to inspect and analyse them with tools like filters and flow graphs

PING COMMAND:

</> Bash

```
ping -c 5 -s 64 google.com
```

USE:

Sends specific number of ping packets (e.g., 5)

OUTPUT:

```
pinging google.com [142.250.183.14] with 64 bytes of data:
Reply from 142.250.183.14: bytes=64 time=18ms TTL=118
Reply from 142.250.183.14: bytes=64 time=19 ms TTL=118
Reply from 142.250.183.14: bytes=64 time=17ms TTL=118
Reply from 142.250.183.14: bytes=64 time=20ms TTL=118
Reply from 142.250.183.14: bytes=64 time=18ms TTL=118

Ping statistics for 142.250.183.14:
    Packets: sent=5, Received=5, Lost=0 (0% loss)
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 20ms, Average = 18ms
```

Exp no: 5 DEVELOP A CUSTOMIZED PING COMMAND
Date: 24/7/26 TO TEST SERVER CONNECTIVITY

AIM:

To develop and execute ping command to test server connectivity and analyse response time and packet loss

THEORY:

1. Why to use Ping command?

It is used to verify whether the server or host is reachable.

2. What is purpose of Ping command?

It is used to verify network connectivity, measure network latency, detect packet loss, troubleshoot network problems

3. How is it established?

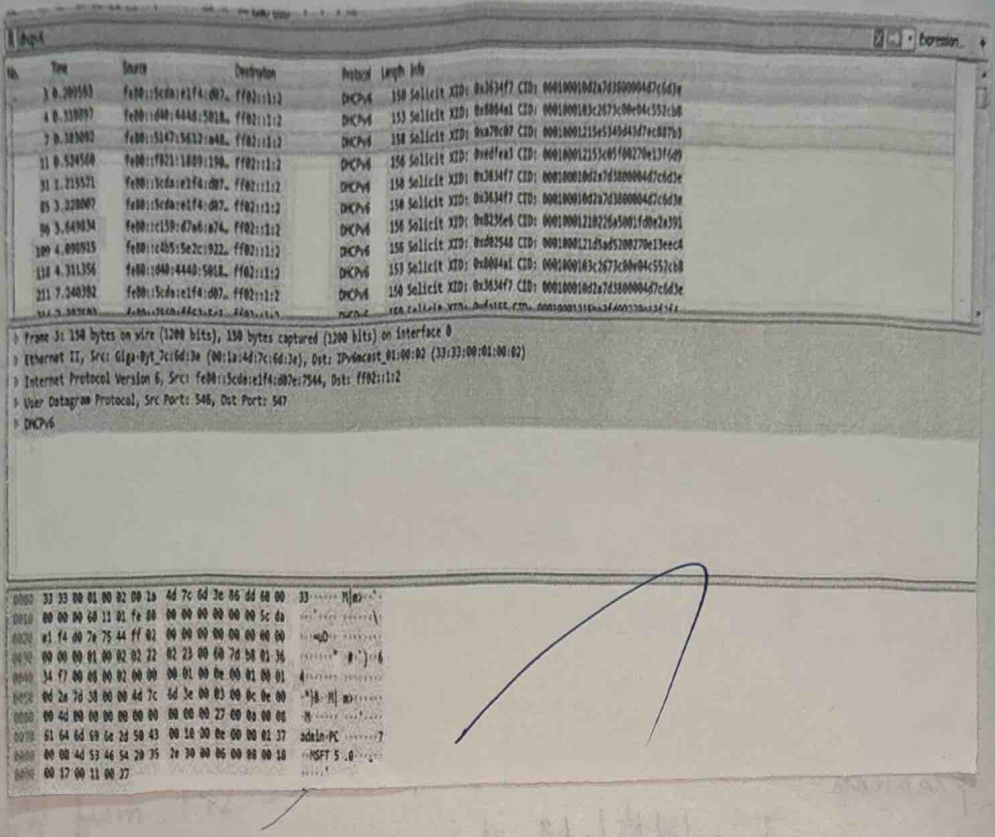
The ping command sends ICMP (Internet control message protocol) echo request packets to the target device. If reachable, it replies with ICMP echo reply packets, confirming establishment

PROCEDURE:

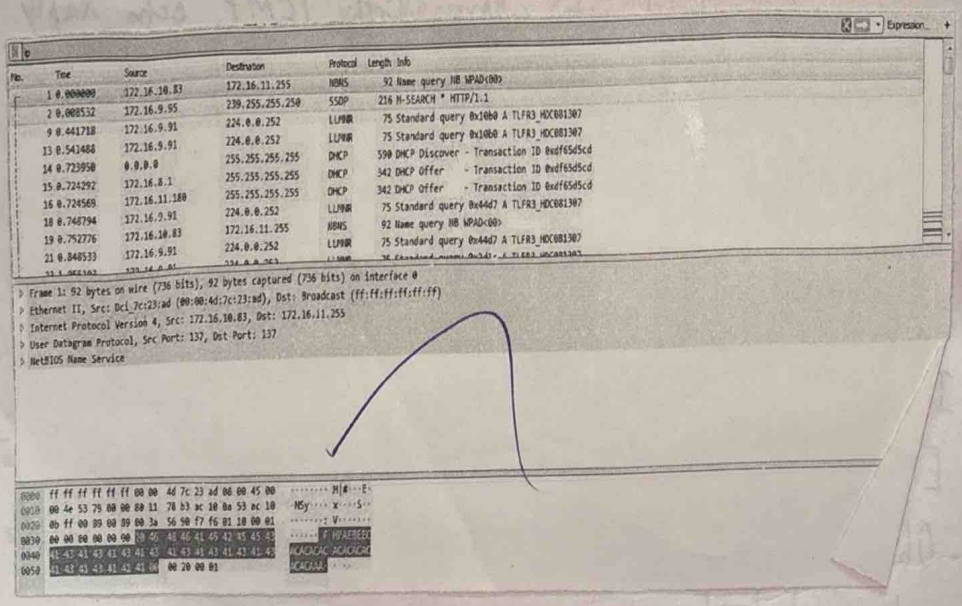
1. Open cmd (windows) or terminal (Linux/macOS)
2. Identify server's IP address or domain name
3. Execute the ping command with the required parameters
4. Observe transmitted and received packets, response time and packet loss
5. Analyse result to determine connectivity

Fourth Experiment:

6.



5.

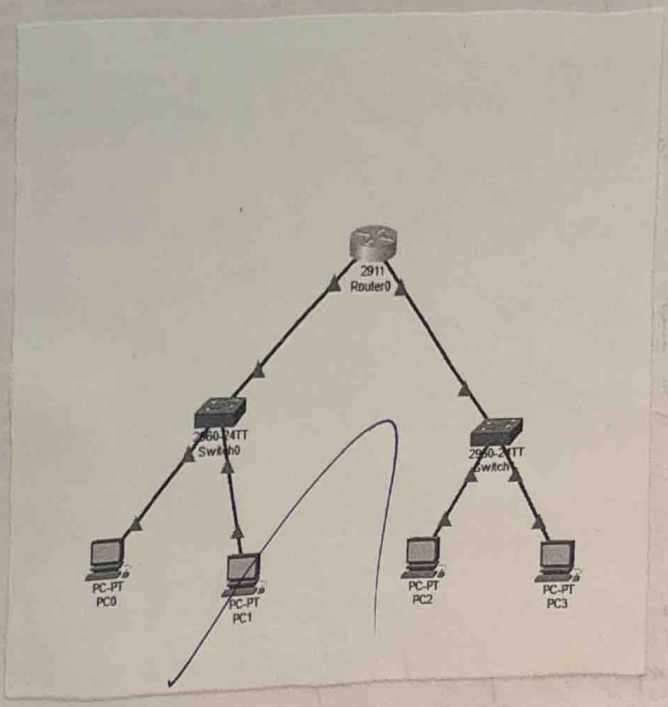


1. Basic Ping Command
   ping www.google.com
2. Ping localhost
   ping localhost
3. Ping using IP Address
   ping 8.8.8.8
4. Continuous ping test command
   ping www.google.com -t
5. Ping with specific no of echo request command
   ping www.google.com -n 5
6. Ping with packet Size
   ping www.google.com -l 1000

OBSERVATION:
- No: Packet no in capture sequence
- Time: Timestamp when packet is captured
- Source: Source IP address of packet
- Destination: Destination IP address of packet
- Protocol: eg: TCP, UDP, HTTP etc.
- Length: size of packet
- Info: additional (like HTTP request/response, ARP request, ICMP echo request)

RESULT:
The ping command was successfully executed. The server responded to all five echo requests with 0% packet loss and an average response time of 18 ms, confirming the stable and active network connectivity

OUTPUT:


2911
Router0
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3

Exp no: 9
Date: 13/8/26

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER

AIM:

To implement classes IP subnetting in CISCO packet tracer and verify communication between different subnets

PROCEDURE:

1. Open cisco packet tracer and create a new network topology
2. Add the required devices to workspace
3. Connect the devices using appropriate cables. Connect PC to switch to router interfaces
4. Use network 192.168.1.0/24 and divide it into smaller subnets using a/27 subnet mask (255.255.224)
5. A /27 subnet provides 30 usable host addresses the required networks are
6. 192.168.1.0/27
7. 192.168.2.0/27
8. 192.168.3.0/27
9. 192.168.4.0/27
10. Configure the router interface through CLI enter

```
enable
configure terminal
interface fast ethernet 0/0
ip address 192.168.1.1.255.255.255.224
no shutdown
exit

interface fast ethernet 0/1
ip address 192.168.21.255.255.255.224
no shutdown
exit
```

Configure the switch ports in access mode:

```
enable
configure terminal
interface fast ethernet 0/1
switch port mode access
exit
interface fast ethernet 0/2
switch port mode access
exit
```

1. Configure each Pc by selecting pc → desktop → ip configuration. Enter appropriate IP address, subnet mask and default gateway.
2. Example IP addressing

Network	Router IP	PC IP range
192.168.10/27	192.168.1.1	192.168.1.11 - 192.168.1.15
192.168.2.0/27	192.168.2.1	192.168.2.11 - 192.168.2.15
192.168.3.0/27	192.168.3.1	192.168.3.11 - 192.168.3.15
192.168.4.0/27	192.168.4.1	192.168.4.11 - 192.168.4.15

1. Configure the remaining router interfaces using appropriate router commands and ensure that interfaces are up
2. Check the status of interfaces using appropriate router commands and ensure that interfaces are up
3. Open the command prompt on PCs and use ping command to test connecting between PCs and their gateway.

OUTPUT:

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=6ms TTL=128
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 3ms

C:\>
```

OBSERVATION:

a. Understanding of subnetting

It is process of dividing large IP network into smaller logical networks. It allows IP address ~~out of~~ to be used efficiently and helps in better network management

b. Advantages

- Efficient utilization of IP addresses
- Reduces network traffic
- Improves network performance
- Makes troubleshooting easier

c. College network

Subnetting is generally used to separate different departments, labs admin and other network segments

RESULT:

Thus, Implementation of subnetting in CISCO packet tracer is executed successfully.

Exp no: 10.a. INTERNETWORKING WITH ROUTERS IN CISCO
Date: 14/8/26 PACKET TRACER SIMULATOR

AIM:

To design and configure a simple internetwork using a router & 2 PCs in CISCO packet tracer and verify connectivity using simple PDU

PROCEDURE:

1. Configure router 1

- open router 1 → CLI & enter:

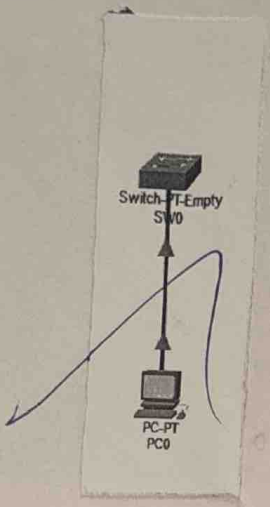
Router > enable
Router # config t
Router (config) # interface
Fast Ethernet 0/0
Router (config-if) # ip address
192.168.10.1 255.255.255.0
Router (config-if) # no
shutdown
Router (config-if) # interface
Fast Ethernet 0/1
Router (config-if) # ip address
192.168.20.1 255.255.255.0
Router (config-if) # no
shutdown

2. Configure PCs

- go to PC → desktop → IP configurations
- config the following

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

OUTPUT:



3. Connect devices

- Connect PC0 FastEthernet 0 to router1 Fa 0/0
- Connect PC1 FastEthernet 0 to router1 Fa 0/1
- use copper straight-through cables
- Send a simple PDU from PC0 to PC1 to verify connectivity

Router configuration:

Device	Fa 0/0 IP	Subnet Mask	Fa 0/1 IP	Subnet Mask
Router1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0

RESULT:

Thus, internetwork was successfully designed and configured. Simple PDU was successfully transferred from PC0 to PC1 confirming successful connectivity

Exp no: 10.b
Date: 14/8/26

INTERNETWORK USING W/RELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

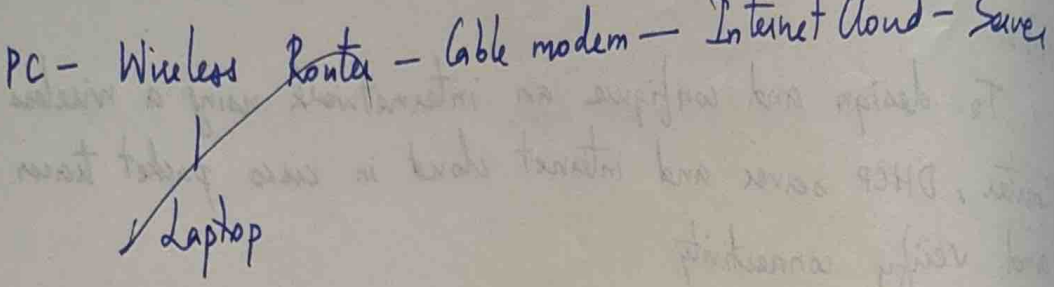
AIM:

To design and configure an internetwork using a wireless Router, DHCP server and internet cloud in cisco packet tracer and verify connectivity

PROCEDURE:

1. Build Network:
   1. Place PC, wireless router, laptop, cable, modem, internet cloud and cisco.com server
   2. Rename devices as required
   3. Connect:
      - PC → wireless router using copper straight-through cable
      - wireless router → Cable Modem using copper straight-through cable
      - cable modem - internet cloud using coaxial cable
      - Internet cloud - cisco.com server using copper straight-through cable
2. Configure wireless router:
   - Set SSID: Home network
   - Enable DHCP server
   - Set DNS Server: 208.67.220.220
   - Save settings
3. Configure Laptop
   - Replace ethernet module with wireless WPC300N
   - Power on laptop
   - Go to desktop → PC wireless
   - Select and connect to Home network

OUTPUT:


PC - Wireless Router - Cable modem — Internet Cloud - Server
Laptop

4. Configure PC
   - Go to desktop → IP configuration
   - Select DHCP
   - use ipconfig/all to verify that PC receives 192.168.0.X IP address
5. Configure Internet Cloud
   - Install PT-cloud-NM-1Cx and PT-cloud-NM-1CFE if required
   - Set ethernet provider network as cable
6. Verify connectivity
   - On PC use:
     ipconfig/release
     ipconfig/renew
   - Verify PC receives 192.168.0.x address
     ping cisco.com
   - Successfully replies confirm connectivity

OBSERVATION:

1. Key feature of wireless router and DHCP
   - Wireless Router: Provides WIFI, SSID, LAN and internet
   - DHCP: Automatically assigns IP address, subnet mask, gateway, ~~DNS~~
2. Significance of DHCP
   - DHCP automatically provides network settings, reduces manual configurations and prevents IP conflicts
3. Internetwork design

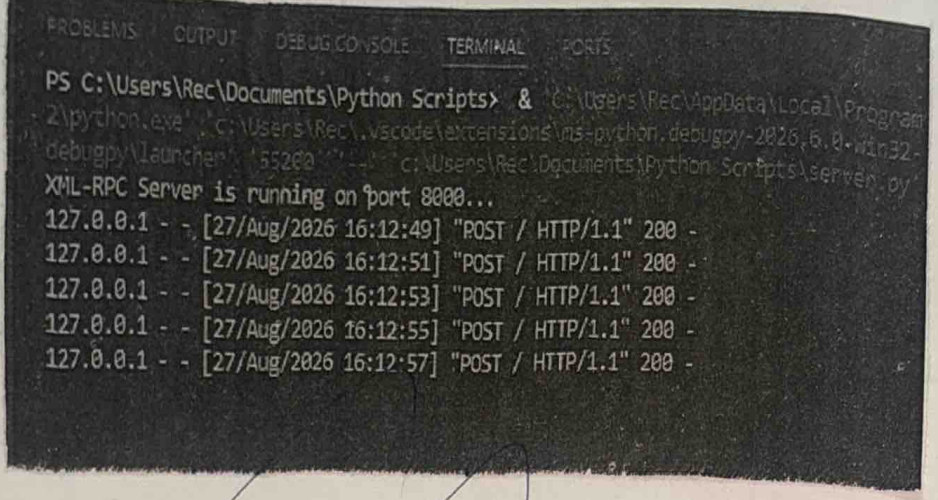
PC — Wireless Router — Cable Modem — Internet Cloud — server
|
Laptop

Device	PC	Wireless Router	Laptop	cisco.com server
IP Address	DHCP (192.168.0.X)	192.168.0.1	DHCP	20.8.67.220.220

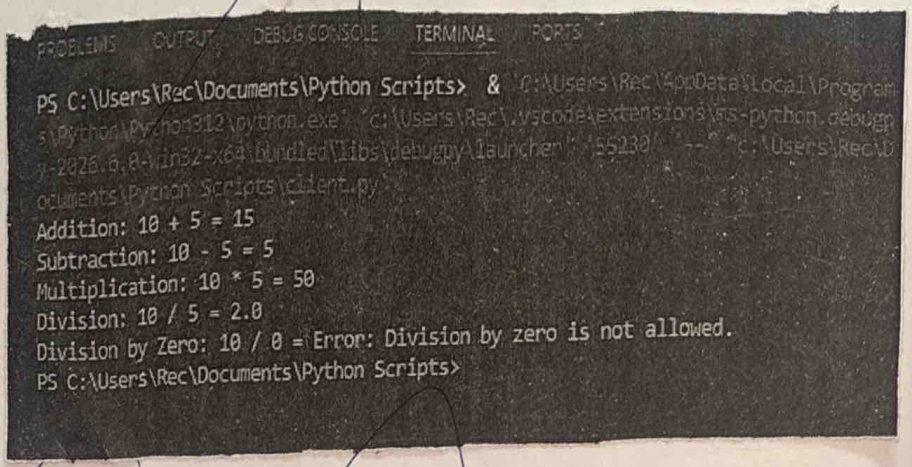
RESULT:

Thus, the internetwork was successfully designed and configured using wireless router, DHCP Server and internet cloud. The PC & Laptop obtained IP address through DHCP. and ping to cisco.com was successful.

Server Output:



Client Output:



Exp no: 7
Date: 2d/8/26

DEVELOP A SIMPLE CALCULATOR USING XMLRPC

AIM:

To develop a simple calculator application using XML-RPC protocol in Python that performs basic arithmetic operations like addition, subtraction, multiplication and division through client-server communication

DESCRIPTION:

XML-RPC is a protocol that uses XML to encode requests and responses and HTTP as transport mechanism. It enables communication between client and server where client invokes methods like hosted on server remotely

- Server hosts arithmetic functions
- Client connects to server to perform operations

CODE:

server.py

```
from xmlrpc.server import SimpleXMLRPCServer

class Calculator:
    def add(self, x, y):
        return x+y
    def subtract(self, x, y):
        return x-y
    def multiply(self, x, y):
        return x*y
    def divide(self, x, y):
        if y!=0:
            return x/y
        else
            return "Error"
server = SimpleXMLRPCServer(("localhost", 8000))
print("XML-RPC Server running on port 8000...")
server.register_instance(Calculator())
server.serve_forever()
```

OUTPUT:

Server Output

XML_RPC Calculator Server Started

Server running on port 8080

Waiting for client

Client Output

Addition: 10+5=15

Subtraction: 10-5=5

Multiplication: 10*5=50

Division: 10/5 = 2.0

Division by Zero: 10/0=

Error: Division by Zero is not allowed

client.py

```
import xmlrpc.client
proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")
print("Addition: 10+5=", proxy.add(10,5))
print("Subtraction: 10-5=", proxy.subtract(10,5))
print("Multiplication: 10*5=", proxy.multiply(10,5))
print("Division: 10/5=", proxy.divide(10,5))
print("Division by zero: 10/0=", proxy.divide(10,0))
```

SOFTWARE REQUIREMENTS:

- OS: Windows/Linux
- Python: Version 3.x or above
- Modules required: xmlrpc.server and xmlrpc.client
- Text editor or IDE: IDLE/VScode/PyCharm

PROCEDURE:

1. Open IDE of python
2. Create 2 separate python files server.py and client.py
3. Write and run server code in server.py
4. Write and run client code in client.py
5. Observe results

RESULT:

XML-RPC calculator is successfully implemented where client communicated with server to perform calculations and displayed accurately.

Exp no: 8
Date: 21/8/26

DEVELOP A PROGRAM TO CREATE REVERSE SHELL USING TCP SOCKETS

AIM:

To develop a program to demonstrate the concept of a reverse shell using TCP sockets in java

THEORY:

A reverse shell is a client-server communication technique in which the client initiates a TCP connection to a listening server. Once the connection is established, the two systems can exchange data through the socket

Java provides serversocket for listening for TCP connection and socket for establishing a connection BufferedReader and PrintWriter are used for receiving and sending data

PROGRAM:

Server.java

```
import java.io.*;
import java.net.*;
Public class Server {
    Public Static void Main(String[] args){
        try{
            ServerSocket Server = new ServerStack(5000);
            System.out.println("Server started");
            System.out.println("Waiting for client...");
            Socket Socket = server.accept();
            System.out.println("Client connected.");
            BufferReader in = new BufferReader(new
                                InputStreamReader(Socket.getInput
                                Stream()));
            PrintWriter out = new PrintWriter(Socket.getOutputStream(),
                                true);
```

OUTPUT:

Server.java

Server started
Waiting for client...
Client connected
Client: Hello from Client;

Client.java
Hello from Client!
Server: Message received by Server

```
String message = in.readline
System.out.println("Client: " + message);
System.out.println("Message received by Server.")
Socket.close();
Server.close();
} catch (IOException e){
    System.out.println("Error: " + e.getMessage));
}
}
}
```

Client.java:

```
import java.io.*
import java.net.*
public static void main (String[] args){
    try{
        Socket socket = new Socket("localhost", 5000);
        BufferedReader in = new BufferedReader(new InputStream
                                Reader(Socket.getInputStream()));
        PrintWriter out = new PrintWriter(socket.getOutputStream(),
                                true);
        System.out.println("Hello from Client!");
        String response = in.readLine();
        System.out.println("Server: " + response);
        Socket.close();
    } Catch (IO Exception e){
        System.out.println("Error:" + e.getMessage());
}
}
}
```

RESULT:

Thus, TCP Client-Server communication program was successfully implemented using Socket and ServerSocket

OUTPUT:

For the scans, Nmap displayed output similar to Starting Nmap

Nmap scan report for 10.48.173.50

Host is up

Nmap done

ARP

Nmap scan report for 10.10.X.1 Host is up

MAC address

Nmap done

ICMP

Nmap scan report for 10.10.X.100 Host is up

Nmap done

Exp no: 12
Date: 29/8/26

NMAP TO DISCOVER LIVE HOSTS USING ARP SCAN, ICMP SCAN AND TCP/UDP PING SCAN IN TRYHACKME PLATFORM

AIM:
To discover whether a target Server/Host is alive or reachable using Nmap with ARP, ICMP, TCP and UDP host-discovery Techniques

THEORY:
Nmap is a network scanning tool used to discover hosts and services on a network. Host discovery determine whether a target system is active or reachable
Different techniques can be used for host discovery:
- ARP Scan (-AR) - Discovers host using ARP requests
- ICMP Scan (-PE) - Uses ICMP Echo requests, Similar to ping
- TCP Ping (-PS) - Uses TCP SYN Probes
- UDP Ping (-PU) - Uses UDP Probes

The -sn option tells Nmap to perform host discovery without port scan

COMMANDS:

1. Basic Host Discovery
```
nmap -sn 10.48.173.50
```
2. ARP Scan
```
sudo nmap -sn -PR 10.48.173.50
```
3. ICMP Scan
```
sudo nmap -sn -PE 10.48.173.50
```
4. TCP Ping Scan
```
sudo nmap -sn -PS80,443 10.48.173.50
```
5. UDP Ping Scan
```
sudo nmap -sn -PU53,161 10.48.173.50
```

RESULT:

Thus, target Server 10.78.173.50 was successfully identified as an active/reachable host using Nmap, ARP, ICMP, TCP and UDP host-discovery technique

SAMPLE OUTPUT:

Case 1: Anonymous login successful

Enter FTP Server address: ftp.example.com

Anonymous FTP login successful

Server allows anonymous access

Case 2: Anonymous login rejected

Enter FTP Server address: ftp.example.com

Anonymous FTP login failed

Reason: 530 login incorrect

Exp no: 6
Date: 27-8-26

BUILDING AN ANONYMOUS FTP SCANNER USING FTPLIB MODULE

AIM:

To develop a python program using ftplib module to scan an authorized FTP server and check whether it allows anonymous login

PROCEDURE:

PROGRAM:

```
from ftplib import FTP, all_errors
host = input("Enter FTP Server address: ")
try:
    ftp = FTP(host, timeout=10)
    ftp.login("anonymous", "anonymous@example.com")
    print("Anonymous FTP login successful")
    print("Server allows anonymous access")
    ftp.quit()
except all_errors as e:
    print("Anonymous FTP login failed")
    print("Reason:", e)
```

OBSERVATION:

1. Python ftplib module was used to establish a connection with FTP server
2. Program attempted to log in using username anonymous
3. If server accepted, angnymous FTP access confirmed
4. If server rejected login/connection failed, an error message was displayed
5. This demonstrates how to identify whether an FTP Server permits anonymous access

RESULT:

Thus, python program was developed successfully using ftplib module

Sample Configuration (PAT):

```
enable
configure terminal
access-list 1 permit 192.168.1.0
0.0.0.255

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit

interface gigabitEthernet 0/1
ip address 203.0.113.1 255.255.255.0
ip nat outside
no shutdown
exit

ip nat inside source list 1
interface gigabitEthernet 0/1
overload
exit
end
```

Exp no: 11
Date: 7-9-26

EXAMINE NETWORK TRANSLATION USING CISCO PACKET TRACER

AIM:

To examine network address translation (NAT) using cisco packet tracer and understand how private IP addresses are translated into public IP address

REQUIREMENTS:

1. Cisco packet tracer
2. One router
3. One switch
4. Two PCs
5. One server or external network for testing

PROCEDURE:

1. Open cisco tracer
2. Add one router, one switch and 2 PCs
3. Connect PC → Switch, switch → router using copper straight through cables
4. Configure PCs with private IP addresses such as 192.168.1.2 and 192.168.1.3
5. Configure an external network and assign IP addresses to router's outside interface
6. Configure an NAT or PAT on router using cisco IP commands
7. Use show ip nat translations to examine translation table

Verification commands:

show ip nat translations

show ip nat statistics

RESULT:

Thus Network Address Translation was examine and translation using PAT was successful.

OUTPUT:

- Network hosts, communication pattern, suspicious activities with true or false value = [redacted]

SAMPLE:

Frame 42: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
Ethernet II, Src: f0:19:af:02:96:f1 (f0:19:af:02:96:f1),
Dst: VMware_f3:ca:52 (00:50:56:f3:ca:52)
Internet Protocol Version 4, Src: 172.16.165.165, Dst: 172.16.165.2
User Datagram protocol, Src Port: 62720, Dst port: 53
Domain Name System (Query)

Exp no: 13
Date: 14-9-26

DEMONSTRATE NETWORK FORENSICS USING PCAPXRAY

AIM:

To perform network forensics using pcapxray tool by analyzing a pcap file and identifying network hosts, communication patterns and suspicious activities

REQUIREMENTS:

1. Computer with python installed
2. PcapXray tool
3. A PCAP file for analysis
4. Wireshark (optional)

PROCEDURE:

1. Download or obtain the Pcapxray tool from its official or trusted repository
2. Install the required dependencies according to its documentation
3. Obtain PCAP file from authorized lab exercise on your own network capture
4. Open Pcapxray tool and load Pcap file
5. Run analysis to generate visualization of network communications
6. Examine the source and destination IP

RESULT:

Thus, the network forensics was demonstrated using pcapxray tool

OUTPUT:

- Server type
- Bandwidth used
- total
- hits
- Unique
- visitors

Top Client/Domain report URL
Apache 12,560 1,240 1.8 GB 192.168.1.10 http://localhost/apache_report/
Nginx 9,840, 980 1.2 GB 192.168.1.20 http://localhost/nginx_report/
Graphical output with monthly summary report

COMMANDS

Step 1: Install

sudo apt-get install webalizer -y

Step 2: Run webalizer - apache log

sudo webalizer /var/log/apache2/access.log -o /var/www/html/apache_report/

Step-3: Run webalizer - nginx log

sudo webalizer /var/log/nginx/access.log -o /var/www/html/nginx_report/

Step-4: Run webalizer - proxy log

sudo webalizer /var/log/squid/access.log -o /var/www/html/proxy_report/

Exp no: 15
Date: 28-9-26

ANALYZE DIFFERENT TYPES OF SERVERS USING WEBALIZER

AIM:

To analyze web server access logs using webalizer tool and study website traffic statistics

REQUIREMENTS:

1. Webalizer software
2. Web server access log files
3. Computer running compatible OS
4. Sample Apache or other supported web server logs

PROCEDURES:

1. Install webalizer and its required dependencies
2. Obtain an access log file from authorized web server or sample log
3. Configure webalizer with correct log file path and report output directory
4. Run webalizer against log file
5. Open generated HTML report in web browser
6. Examine visitor stats, page requests, URLs and data transfer info

Sample command:

On Linux system with webalizer installed

webalizer /var/log/apache2/access.log

RESULT:

Thus web server access logs were analyzed using webalizer and website traffic statistics were examined through generated reports